The Definition of the Derivative

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Use the Second Derivative Test to classify the critical points of $f(x) = 3x^3 - 4x$.

$f'(x) = 3(3)x^2 - 4 = 0 \Rightarrow x = \pm\frac{2}{3}$. Since $f''(x) = 18x$, $f''(-\frac{2}{3}) < 0$ (local maximum) and $f''(\frac{2}{3}) > 0$ (local minimum). Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

2 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The graph of the function $f(x) = ax - 3$ is shown below. The area of the shaded region between the graph of $f(x)$ and the x-axis from $x = 0$ to $x = 4$ is 20 square units. Find the value of the parameter a.

The function is given by $f(x) = ax - 3$. The area of the shaded region is represented by the definite integral of $f(x)$ from $x = 0$ to $x = 4$. We are given that this area is 20. $\int_0^4 (ax - 3)\,dx = 20$ First, find the antiderivative of $ax - 3$: $\int (ax - 3)\,dx = \frac{a}{2}x^2 - 3x + C$ Now, evaluate the definite integral from 0 to 4: $\left[\frac{a}{2}x^2 - 3x\right]_0^4 = \left(\frac{a}{2}(4)^2 - 3(4)\right) - \left(\frac{a}{2}(0)^2 - 3(0)\right)$.

3 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The velocity of a particle moving along a straight line is given by the function $v(t) = t^2 - 6t + 8$ m/s. What is the instantaneous acceleration of the particle at time $t = 3$ seconds?

The velocity of the particle is given by $v(t) = t^2 - 6t + 8$. Acceleration is the rate of change of velocity with respect to time, which means acceleration is the derivative of the velocity function, $a(t) = v'(t)$. Differentiating $v(t)$ with respect to t, we get: $a(t) = \frac{d}{dt}(t^2 - 6t + 8) = 2t - 6$. We need to find the instantaneous acceleration at time $t = 3$ seconds. Substituting $t = 3$ into the acceleration function: $a(3) = 2(3) - 6 = 6 - 6 = 0$ m/s^2. Therefore, the instantaneous acceleratio.

4 **[Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Use L'Hopital's Rule to evaluate $\lim_{x\to\infty} \frac{7x}{e^{4x}}$.

This is $\frac{\infty}{\infty}$ form. Applying L'Hopital's Rule: $\lim_{x\to\infty} \frac{7}{4e^{4x}} = 0$, since the exponential grows without bound. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

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5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A 5-ft ladder leans against a wall. The bottom slides away from the wall at 2 ft/s. Find the rate at which the top of the ladder is sliding down the wall when the bottom is 3 ft from the wall.

$x^2 + y^2 = 5^2$. At $x = 3$, $y = 4$. Differentiating: $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt} = -\frac{3}{2}$ ft/s. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the area of the region bounded by $y = 28x$ and $y = x^2$ in the first quadrant.

Intersections at $x = 0$ and $x = 28$. Area $= \int_0^{28} (28x - x^2)\,dx = \left[\frac{28}{2}x^2 - \frac{x^3}{3}\right]_0^{28} = \frac{21952}{6}$. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A differentiable function $f(x)$ is given by the equation $f(x) = x^2 - 6x + 10$. Find the instantaneous rate of change of $f(x)$ at $x = 4$.

The instantaneous rate of change of a function at a point is given by its derivative evaluated at that point. First, we find the derivative of $f(x) = x^2 - 6x + 10$. Using the power rule, $f'(x) = 2x - 6$. Next, we evaluate the derivative at $x = 4$: $f'(4) = 2(4) - 6 = 8 - 6 = 2$. Therefore, the instantaneous rate of change of $f(x)$ at $x = 4$ is 2.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 1 · FoundationsFind the derivative of $f(x) = x^4e^x - \frac{\sin x}{x^2+1}$.

Apply Product Rule to x^4e^x and Quotient Rule to $\frac{\sin x}{x^2+1}$: $\frac{d}{dx}[x^4e^x] = 4x^3e^x + x^4e^x = x^3e^x(4+x)$. $\frac{d}{dx}\left[\frac{\sin x}{x^2+1}\right] = \frac{(\cos x)(x^2+1) - (\sin x)(2x)}{(x^2+1)^2} = \frac{(x^2+1)\cos x - 2x\sin x}{(x^2+1)^2}$. Combining the results: $f'(x) = \mathbf{x^3e^x(4+x) - \frac{(x^2+1)\cos x - 2x\sin x}{(x^2+1)^2}}$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 3 · AdvancedFind the 42nd derivative $f^{(42)}(x)$ of $f(x) = \sin(2x)$.

Each differentiation introduces a factor of 2 by the Chain Rule and cycles through sine and cosine with period 4: $f'(x) = 2\cos(2x)$ $f''(x) = -2^2\sin(2x)$ $f'''(x) = -2^3\cos(2x)$ $f^{(4)}(x) = 2^4\sin(2x)$ Divide order by 4: $42 = 4(10) + 2$, so $f^{(42)}$ has the same trigonometric phase as $f''(x)$:

$f^{(42)}(x) = -2^{42}\sin(2x) = \mathbf{-4{,}398{,}046{,}511{,}104}\sin\mathbf{(2x)}$.

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10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Level 2 · MediumLet f be a function whose derivative is $f'(x) = (x-1)^2(x+2)(x-4)$. Determine the x-coordinates of all relative extrema of f and classify each with justification.

Critical numbers occur where $f'(x) = 0$: $x = -2$, $x = 1$, $x = 4$. Test intervals of $f'(x)$:On $(-\infty, -2)$: $(-) \times (+) \times (-) = +$ ($f' > 0$).On $(-2, 1)$: $(+) \times (+) \times (-) = -$ (x.On $(1, 4)$: $(+) \times (+) \times (-) = -$ (x.On $(4, \infty)$: $(+) \times (+) \times (+) = +$ ($f' > 0$). Since f' changes from negative to positive, f has a relative minimum at $x = 4$.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Level 2 · MediumThe base of a solid is bounded by $y = \sqrt{x}$, the line $x = 9$, and the x-axis. Cross sections perpendicular to the x-axis are equilateral triangles. Find the volume of the solid.

Base side length: $s(x) = \sqrt{x} - 0 = \sqrt{x}$. Equilateral triangle area formula: $A(x) = \frac{\sqrt{3}}{4}[s(x)]^2 = \frac{\sqrt{3}}{4}(\sqrt{x})^2 = \frac{\sqrt{3}}{4}x$ Integrate from $x = 0$ to $x = 9$: $V = \int_0^9 \frac{\sqrt{3}}{4}x\,dx = \frac{\sqrt{3}}{4}\left[\frac{x^2}{2}\right]_0^9 = \frac{\sqrt{3}}{4} \cdot \frac{81}{2} = \mathbf{\frac{81\sqrt{3}}{8}} \approx 17.537$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Level 2 · MediumFind the value of a and b such that the line $y = 4x - 1$ is tangent to $f(x) = ax^2 + bx$ at $x = 1$.

1. Value match at $x = 1$: The tangent line gives $y(1) = 4(1) - 1 = 3$. Thus $f(1) = a(1)^2 + b(1) = 3 \Rightarrow a + b = 3$. 2. Slope match at $x = 1$: The tangent line has slope $m = 4$. Derivative: $f'(x) = 2ax + b$. Thus $f'(1) = 2a + b = 4$. 3. Solve system: Subtract equations: $(2a + b) - (a + b) = 4 - 3 \Rightarrow \mathbf{a = 1}$. Then $b = 3 - a = \mathbf{2}$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Level 2 · MediumFind the derivative $h'(x)$ for the accumulation function: $h(x) = \int_{x^2}^{\sin(3x)} \sqrt{1+t^4}\,dt$

Apply the Leibniz Rule with the Chain Rule: $h'(x) = \sqrt{1+(\sin(3x))^4} \cdot \frac{d}{dx}[\sin(3x)] - \sqrt{1+(x^2)^4} \cdot \frac{d}{dx}[x^2]$ Differentiating the boundary functions: $\frac{d}{dx}[\sin(3x)] = 3\cos(3x)$, $\frac{d}{dx}[x^2] = 2x$ Combining terms yields: $h'(x) = \mathbf{3}\cos\mathbf{(3x)}\sqrt{\mathbf{1+\sin^4(3x)}} \mathbf{- 2x}\sqrt{\mathbf{1+x^8}}$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Level 3 · AdvancedIf a student earns 32 out of 45 points on Section I and 34 out of 54 points on Section II, what is their composite score out of 108?

Multiple Choice is scaled by 1.2: $32 \times 1.2 = 38.4$. Free Response points count 1:1: $34 \times 1 = 34$. Total composite score: $38.4 + 34 = \mathbf{72.4}$ out of 108 points, which safely earns an AP score of 5.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives**

[AP Calculus AB Exam Pacing & Scoring Strategy · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 3.01 The Chain Rule**

[AP Calculus AB Exam Format & Units Overview · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions**

[Unit 2 · Differentiation — Definition and Fundamental Properties · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion**

Level 1 · FoundationsWhich single unit on the AP Calculus BC exam carries the highest individual weighting, and what specific topic within it is virtually guaranteed to appear on Section II as Free Response Question 6?

Unit 10 (Infinite Sequences and Series) carries the highest individual weighting at 17–18% of the total exam. On Section II Part B, Question 6 is historically and systematically dedicated to Taylor and Maclaurin polynomials, series representations, radius/interval of convergence, and error bounds (Lagrange or Alternating Series Error Bound).

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule**

Level 3 · AdvancedEvaluate the limit: $\lim_{x \to 0^+} (\cos x)^{1/x^2}$

This is an indeterminate power form of type 1^∞. Let $y = (\cos x)^{1/x^2}$. Take the natural logarithm: $\ln y = \ln\left((\cos x)^{1/x^2}\right) = \frac{\ln(\cos x)}{x^2}$ Evaluate $\lim_{x \to 0^+} \ln y = \lim_{x \to 0^+} \frac{\ln(\cos x)}{x^2}$. Check limits separately: $\lim_{x \to 0^+} \ln(\cos x) = \ln 1 = 0$ and $\lim_{x \to 0^+} x^2 = 0$. Apply L'Hôpital's Rule.

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20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Level 1 · FoundationsFind the area of the region bounded by $x = y^2$ and $x = y + 2$ using integration with respect to y.

Find intersections in terms of y: $y^2 = y + 2 \Rightarrow y^2 - y - 2 = 0 \Rightarrow (y-2)(y+1) = 0 \Rightarrow y = -1,\ y = 2$ For $y \in (-1, 2)$, test $y = 0$: $x_{\text{line}} = 2 > x_{\text{parabola}} = 0$. The right curve is $x = y + 2$ and the left curve is $x = y^2$. Integrate with respect to y: $A = \int_{-1}^{2} [(y+2) - y^2]\, dy = \left[\frac{y^2}{2} + 2y - \frac{y^3}{3}\right]_{-1}^{2}$ At $y = 2$: $2 + 4 - \frac{8}{3} = 6 - \frac{8}{3} = \frac{10}{3}$. At $y = -1$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which limit expression represents the derivative $f'(x)$ by definition?

a. $\lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ ✓

b. $\lim_{h \to 0} \frac{f(x+h) + f(x)}{h}$

c. $\lim_{h \to \infty} \frac{f(x+h) - f(x)}{h}$

d. $\frac{f(x+h) - f(x)}{h}$

By definition, $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which function and value of c are represented by $\lim_{x \to 4} \frac{\sqrt{x} - 2}{x - 4}$?

a. $f(x) = \sqrt{x}$ **at** $c = 4$ ✓

b. $f(x) = x^2$ at $c = 2$

c. $f(x) = \sqrt{x}$ at $c = 2$

d. $f(x) = \frac{1}{\sqrt{x}}$ at $c = 4$

Using the alternate form $f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}$, this corresponds to $f(x) = \sqrt{x}$ at $c = 4$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If a function f is differentiable at $x = c$, which of the following statements MUST be true?

a. **f is continuous at $x = c$** ✓

b. $f'(c) > 0$

c. $f''(c)$ exists

d. f has a local extremum at $x = c$

Differentiability implies continuity. (The converse is false, as demonstrated by sharp corners like $f(x) = |x|$).

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Why is the function $f(x) = |x - 3|$ not differentiable at $x = 3$?

a. **The left-hand and right-hand derivative limits are unequal (corner)** ✓

b. $f(3)$ is undefined

c. f is not continuous at $x = 3$

d. $\lim_{x \to 3} f(x) = \infty$

For $x < 3$, $f'(x) = -1$, and for $x > 3$, $f'(x) = 1$. The corner creates unequal one-sided derivative limits.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The graph of $f(x) = x^{1/3}$ has what feature at $x = 0$ that prevents differentiability?

a. **A vertical tangent line where $f'(x) \to \infty$** ✓

b. A jump discontinuity

c. A sharp cusp

d. A hole

$f'(x) = \frac{1}{3}x^{-2/3} = \frac{1}{3x^{2/3}}$. As $x \to 0$, $f'(x) \to \infty$, which corresponds to a vertical tangent line.

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the equation of the tangent line to $f(x) = x^3 - 2x$ at the point $(2, 4)$?

a. $y - 4 = 10(x - 2)$ ✓

b. $y - 4 = 8(x - 2)$

c. $y - 4 = 12(x - 2)$

d. $y = 10x - 12$

$f'(x) = 3x^2 - 2$. Slope at $x = 2$ is $f'(2) = 3(4) - 2 = 10$. In point-slope form: $y - 4 = 10(x - 2)$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the slope of the normal line to $y = x^2 + 3x$ at $x = 1$?

a. $-\frac{1}{5}$ ✓

b. 5

c. -5

d. $\frac{1}{5}$

$y' = 2x + 3 \Rightarrow y'(1) = 5$. The normal line is perpendicular to the tangent line, so its slope is $-\frac{1}{5}$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

At what point on the parabola $y = x^2 - 4x + 3$ is the tangent line horizontal?

a. $(2, -1)$ ✓

b. $(2, 0)$

c. $(4, 3)$

d. $(0, 3)$

Horizontal tangent requires $y' = 0$: $2x - 4 = 0 \Rightarrow x = 2$. $y(2) = 4 - 8 + 3 = -1$. Point is $(2, -1)$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Given values $t = 2, 4, 6$ with $s(2) = 14, s(4) = 30, s(6) = 54$, estimate $s'(4)$ using the average rate of change over $[2, 6]$.

a. 10 ✓

b. 8

c. 16

d. 12

Symmetric difference quotient: $s'(4) \approx \frac{s(6) - s(2)}{6 - 2} = \frac{54 - 14}{4} = \frac{40}{4} = 10$.

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If the graph of f has a local maximum at $x = 5$ and is differentiable, what is $f'(5)$?

a. 0 ✓

b. 1

c. Positive

d. Undefined

By Fermat's theorem, the derivative at an interior local extremum of a differentiable function is zero: $f'(5) = 0$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If $f'(x) > 0$ for all $x \in (a, b)$, what does this indicate about the graph of f on (a, b)?

a. f **is strictly increasing** ✓

b. f is strictly decreasing

c. f is concave up

d. f is linear

A strictly positive first derivative implies that the function is strictly increasing on the interval.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A region R is bounded by the curve $y = \frac{1}{x^2+1}$, the x-axis, and the vertical lines $x = 1$ and $x = 3$. A solid is formed by revolving this region R about the y-axis. Find the volume of this solid. \\ \\ \\ (Hint: Consider the shell method and use a substitution that relates to the derivative of the denominator of the integrand.)

a. $\pi \ln(10)$

b. $2\pi \ln(5)$

c. $\pi \ln(5)$ ✓

d. $2\pi \ln(10)$

The volume of the solid generated by revolving the region bounded by $y = f(x)$, the x-axis, and $x = a, x = b$ about the y-axis can be found using the shell method. The formula for the volume is $V = \int_a^b 2\pi x f(x)\,dx$. \\ In this case, $f(x) = \frac{1}{x^2+1}$, $a = 1$, and $b = 3$. \\ So, the volume is $V = \int_1^3 2\pi x\left(\frac{1}{x^2+1}\right)dx$. \\ We can pull out the constant 2π: $V = 2\pi\int_1^3 \frac{x}{x^2+1}\,dx$. \\ To evaluate this integral, we use a substituti.

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33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The slope of the tangent to $y = x^3$ at $x = 2$ is:

a. 6

b. 8

c. **12** ✓

d. 16

Applying Unit 2 · Differentiation — Definition and Fundamental Properties principles, $y' = 3x^2$; at $x = 2$: $3(4) = 12$ Therefore, the verified correct option is 12.

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find the derivative of $f(x) = 4e^x$.

a. e^x

b. $4e^x$ ✓

c. $-4e^x$

d. $4e^x + 4$

Applying Unit 2 · Differentiation — Definition and Fundamental Properties principles, $f'(x) = 4e^x$. Therefore, the verified correct option is $4e^x$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The table shows values of $g(x)$ at $x = [0, 3, 6, 8]$: $g = [2, 3, 8, 7]$. Using a right Riemann sum with these subintervals, approximate $\int_0^8 g(x)\,dx$.

a. 41

b. **47** ✓

c. 31

d. 50

Sum of (width × right height) over each subinterval: $3(3) + 3(8) + 2(7) = 47$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The table shows values of $g(x)$ at $x = [0, 2, 5, 7]$: $g = [7, 8, 12, 4]$. Using a left Riemann sum with these subintervals, approximate $\int_0^7 g(x)\,dx$.

a. 56

b. 65

c. **62** ✓

d. 60

Sum of (width × left height) over each subinterval: $2(7) + 3(8) + 2(12) = 62$.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Using a left Riemann sum with 4 subintervals of equal width on $[0, 4]$, approximate $\int_0^4 (-x^2 + x + 4)\,dx$.

a. 5

b. -4

c. 2

d. **8** ✓

$\Delta x = 1$; heights at the left endpoints give $1(4 + 4 + 2 - 2) = 8$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Using a midpoint Riemann sum with 4 subintervals on $[0, 8]$, approximate $\int_0^8 (-2x^2 + 1)\,dx$.

a. -333

b. -472

c. -326

d. **-328** ✓

Midpoints at $\Delta x = 2$ intervals give $2(-1 - 17 - 49 - 97) = -328$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

The limit $\lim_{n\to\infty} \sum_{i=1}^{n} 3\left(\frac{4i}{n}\right)^2 \frac{4}{n}$ equals which definite integral's value?

a. 128

b. **64** ✓

c. 61

d. 32

This is a right Riemann sum for $\int_0^4 3x^2\,dx = 64$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Using a right Riemann sum with 4 subintervals of equal width, approximate $\int_0^4 (x + 2)\,dx$.

a. 16

b. 14

c. 20

d. **18** ✓

With $\Delta x = 1$, the right sum is $1(3 + 4 + 5 + 6) = 18$.

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Using a left Riemann sum with 2 subintervals of equal width, approximate $\int_0^6 (3x)\,dx$.

a. 81

b. 9

c. 23

d. **27** ✓

With $\Delta x = 3$, the left sum is $3(0 + 9) = 27$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

A plane region R is bounded by the graph of $y = e^{-x^2}$, the x-axis, and the lines $x = 0$ and $x = 1$. A solid is generated by revolving the region R about the line $y = -2$. The cross-sections of the solid perpendicular to the x-axis are squares. The volume of this solid is given by $V = \int_a^b A(x)dx$. (a) Find the outer radius and inner radius of a representative disk if the region were revolved about the line $y = -2$ using the washer method. (b) Set up, but do not evaluate, an integral expression for the volume of the solid generated by revolving the region R about the line $y = -2$ using the washer method. (c) Set up, but do not evaluate, an integral expression for the volume of the solid generated by taking cross-sections perpendicular to the x-axis that are squares. (d) Explain why the integral expression in part (c) is the correct representation for the volume of the solid.

a. The radius of a disk in the washer method is $R(x) = e^{-x^2} + 2$ and $r(x) = 2$.

b. The volume using the washer method is $V = \int_0^1 \pi((e^{-x^2} + 2)^2 - 2^2)dx$.

c. The area of a square cross-section is $A(x) = (e^{-x^2})^2$.

d. **The integral expression in part (c) represents the volume because it sums the areas of square cross-sections perpendicular to the x-axis over the interval [0, 1].** ✓

(a) When revolving the region R about the line $y = -2$, the outer radius $R(x)$ is the distance from $y = -2$ to the upper boundary of the region $y = e^{-x^2}$, which is $R(x) = e^{-x^2} - (-2) = e^{-x^2} + 2$. The inner radius $r(x)$ is the distance from $y = -2$ to the lower boundary of the region (the x-axis, $y = 0$), which is $r(x) = 0 - (-2) = 2$. (b) Using the washer method, the volume V is given by the integral of the area of the washers, which is $\pi(R(x)^2 - r(x)^2)$. Therefore, the integral express.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Consider a region R in the xy-plane bounded by the graphs of $y = \frac{1}{2}x^2$ and $y = 2x$. A solid has its base in region R, and for each value of x in the interval $[0, 4]$, the cross-section perpendicular to the x-axis is an equilateral triangle with side length $s(x) = 2x - \frac{1}{2}x^2$. Write an integral expression that represents the volume of this solid. You do not need to evaluate the integral.

a. $V = \int_0^4 \frac{\sqrt{3}}{4}\left(2x - \frac{1}{2}x^2\right)^2 dx$ ✓

b. $V = \int_0^4 \frac{\sqrt{3}}{4}\left(2x - \frac{1}{2}x^2\right) dx$

c. $V = \int_0^4 \frac{1}{2}\left(2x - \frac{1}{2}x^2\right)^2 dx$

d. $V = \int_0^4 \frac{\sqrt{3}}{2}\left(2x - \frac{1}{2}x^2\right) dx$

The base of the solid is defined by the region R bounded by $y = \frac{1}{2}x^2$ and $y = 2x$. The upper bound is $y = 2x$ and the lower bound is $y = \frac{1}{2}x^2$. The intersection points are found by setting $\frac{1}{2}x^2 = 2x$, which gives $x^2 = 4x$, so $x^2 - 4x = 0$, and $x(x - 4) = 0$. Thus, the region extends from $x = 0$ to $x = 4$. The side length of the equilateral triangle cross-section is given by $s(x) = 2x - \frac{1}{2}x^2$. The area of an equilateral triangle with side length s is.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Consider the function $f(x)$ whose derivative is given by $f'(x) = \frac{\sin(x)}{x^2 + 1}$. A related function $g(x)$ is defined by $g(x) = \int_0^x f(t)dt$. If $f(0) = k$, where k is a positive constant, find the value of $g'(0)$ and determine if $g(x)$ is increasing or decreasing at $x = 0$.

a. $g'(0) = k$**, and** $g(x)$ **is increasing.** ✓

b. $g'(0) = 0$, and $g(x)$ is neither increasing nor decreasing.

c. $g'(0) = 1$, and $g(x)$ is increasing.

d. $g'(0) = k$, and $g(x)$ is decreasing.

We are given $g(x) = \int_0^x f(t)dt$. By the Fundamental Theorem of Calculus Part 1, the derivative of $g(x)$ is $g'(x) = f(x)$. We need to find $g'(0)$. Substituting $x = 0$ into $g'(x)$, we get $g'(0) = f(0)$. We are given that $f(0) = k$, where k is a positive constant. Therefore, $g'(0) = k$. To determine if $g(x)$ is increasing or decreasing at $x = 0$, we examine the sign of $g'(0)$. Since k is a positive constant, $g'(0) = k > 0$. A function is increasing at a point if its derivative at that point is posit.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Area Between Curves

The graph of a differentiable function f is shown below. The function f has a local maximum at $x = a$ and a local minimum at $x = b$. The tangent line to the graph of f at $x = c$ has the equation $y - f(c) = \frac{1}{2}(x - c)$. Given that $f(c) = 3$ and $f'(c) = \frac{1}{2}$, find the value of c.

a. 4

b. 3

c. **2** ✓

d. 1

The equation of the tangent line to the graph of a function f at $x = c$ is given by $y - f(c) = f'(c)(x - c)$. We are given that the equation of the tangent line at $x = c$ is $y - f(c) = \frac{1}{2}(x - c)$. Comparing this to the general form, we see that $f'(c) = \frac{1}{2}$. We are also given that $f(c) = 3$. The graph shows a point on the function $f(x)$ at $(x, y) = (2, 3)$. Since the point $(c, f(c))$ must lie on the graph of f, we have $c = 2$ and $f(c) = 3$. Therefore, the value of c is 2.

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